

Gaussian Dust and Christoffel Recovery in Infinite Uniform Convolutions

Exact non-estimability, a complete compact-class criterion,
and geometric moment formulas beyond finite capacity

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Abstract

The finite-capacity inverse problems in ProveIt’s Fabius–Rvachev research program change qualitatively when the number of uniform convolution factors is unrestricted. We study a known smooth compactly supported background plus an independent Gaussian and any square-summable sequence of centered uniform factors. The underlying distributional compactification and identifiability belong to the existing DUSTPAN theory of Billey–Swanson; they are credited and re-established in a variance-weighted moment representation suited to inference. In this experiment the Gaussian variance is identifiable, but its exact minimax absolute-error risk is $V/2$, and its exact squared-error risk is $V^2/4$, at every finite sample size when total latent variance is at most V . On a compact subclass, uniform consistency is possible if and only if the uniform-factor energy tails vanish uniformly. We provide finite-cumulant upper certificates, a Christoffel hierarchy with an explicit tail bound, and a rank-free total-variation expansion quantifying Gaussian approximation. For squared scales Cr^j , the degree- d Christoffel bias is exactly $C(1-r)r^d/[3(1-r^{d+1})^2]$. Coefficient-constrained regularization gives noise bounds and statistical upper rates on two-sided polynomial and geometric envelope classes. The article distinguishes classical tools, derived results, and unresolved minimax and computational questions. No proof-assistant verification or worldwide priority claim is made.

Keywords. Fabius function; Rvachev up-function; DUSTPAN distributions; Gaussian component; infinite convolution; Christoffel function; cumulants; statistical identifiability; uniform consistency.

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1 Research question, provenance, and main conclusions

1.1 What is being extended

ProveIt contains exact Fabius–Rvachev arithmetic and several recent inverse problem reports. The finite-factor stability report studies a fixed number of unknown uniform factors against a known smooth background. Its successor allows an unknown Gaussian variance but keeps the capacity fixed. The earlier infinite-spectrum report constructs moment-preserving perturbations and quantitative inverse-instability examples near the dyadic spectrum [1, 2, 3, 4]. These are different questions: neither finite-capacity rates nor full-law identifiability alone determine what happens to Gaussian-variance estimation when arbitrarily many tiny factors are allowed.

We address the following specific extension:

For an unrestricted square-summable uniform spectrum, can one distinguish actual Gaussian variance from the accumulated variance of unresolved small uniform factors? Which restrictions make this distinction statistically recoverable, and can the recovery be certified from finitely many cumulants?

The answer has both a negative and a constructive part. The unrestricted statistical problem has exactly the risk of ignoring the data. On compact subclasses, one precise tail condition is necessary and sufficient for uniform consistency. Positive polynomial certificates turn this qualitative condition into an explicit method.

1.2 Classical results that are not claimed as new

Billey and Swanson introduced DUSTPAN distributions, characterized their compact parameter space, and proved the closure and identifiability statements for generalized uniform sums plus a normal component [6, Theorem 1.15, Theorem 3.13, Corollaries 3.28–3.29]. Their interval lengths are twice our half-lengths. Thus the existence of a Gaussian component in the closure, the uniqueness of the spectrum, and the basic compactification are *not* discoveries of this article.

The variational definition and moment-matrix formula for Christoffel functions are also classical; [7, Section 2.2] gives a modern account, including singular moment matrices. Our geometric computation is an elementary Cauchy-determinant specialization, related to the geometric orthogonality measures in the little q -Jacobi family [10, Section 18.27(iv)]. No novelty is asserted for those identities or for general compact moment determinacy. The distinction between exact moment determination and unstable inversion has an extensive literature; [8] treats a different, operator-theoretic Hausdorff moment setting.

ProveIt also already contains Christoffel reconstruction of the physical measure $\text{up}(x) \, dx$, as recorded in its representation-frontier corpus audit [5]. Our measure is instead $3s\delta_0 + \sum_j a_j^2 \delta_{a_j^2}$: its atoms encode convolution factors, not spatial observations of the up-density. The distinction prevents conflating an existing density-reconstruction result with the variance-recovery problem treated here.

The contribution developed here is the combination of a variance-weighted spectral measure with the smoothed statistical experiment: exact finite-sample risk, the compact-class equivalence theorem, explicit rank-free L^1 remainder bounds, and constructive Gaussian-variance certificates with exact geometric bias and controlled noisy implementations. These results have full proofs below. Their worldwide publication priority has not been established.

1.3 A guide to the theorems

Fix a variance budget $V > 0$. In the notation introduced in section 2, s is Gaussian variance, v is *total latent variance*, and a_j are uniform half-lengths.

Question	Answer proved here or credited below
What does the exact law identify?	All a_j , with multiplicity, and s ; the underlying identification is classical. The spectral measure in theorem 3.2 makes the Gaussian variance an atom at zero.
What can a finite sample learn without tail restrictions?	Exactly minimax risk $V/2$ in absolute error and $V^2/4$ in squared error, theorem 6.1.
Which compact classes are learnable?	Exactly those with uniformly vanishing ℓ^2 tails of a , theorem 8.1.
Can finite cumulants certify recovery?	Yes: decreasing positive certificates and a Christoffel hierarchy, theorems 7.1 and 9.2.
What happens for geometric factors?	An exact finite-degree bias formula, theorem 10.1; the dyadic specialization is displayed explicitly.
How rapidly can tiny factors imitate Gaussian noise?	A total-variation asymptotic with an explicit uniform remainder, theorems 5.1 and 5.2.

1.4 Snapshot and verification boundary

The inspected repository snapshot is commit

458ccfb80b9940220819091da107e82bb970dcca.

The relevant directory is

Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/inverse-and-sampling/.

The accompanying source audit records the particular files inspected; this is not an audit of every theorem in ProveIt. None of the proofs below assumes that an unreviewed repository theorem is correct. Supporting computations check finite algebraic identities, not the universal analytic or statistical claims. The manuscript has not been independently refereed or checked in Lean or Rocq. No repository files were changed.

2 Model and elementary analytic facts

Let $U_1, U_2, \dots$ be independent uniform random variables on $[-1, 1]$, and let G be an independent standard normal. Let B be independent of them, known, centered, and supported in $[-H, H]$, with density $b \in C_c^\infty(\mathbb{R})$. Centering a known background causes no loss of generality. The statistical observations are independent copies of

$$Y = B + Z, \quad Z = \sqrt{s}G + \sum_{j \geq 1} a_j U_j, \quad a_1 \geq a_2 \geq \dots \geq 0, \quad \sum_{j \geq 1} a_j^2 < \infty. \quad (2.1)$$

The series converges in L^2 and almost surely, by independence, centering, and the summability of the variances. The L^2 statement follows directly from the Cauchy criterion; the almost-sure statement is Kolmogorov's convergence theorem, also recalled in [6, Theorem 3.9]. Only the L^2 realization is needed in the subsequent arguments.

Write

$$x_j = a_j^2, \quad p_k(a) = \sum_{j \geq 1} x_j^k, \quad v = s + \frac{p_1(a)}{3}, \quad 0 \leq v \leq V, \quad L = 3V. \quad (2.2)$$

Thus s and v are variances, not standard deviations. The latent law is $\mu_{v,a} = \mathcal{L}(Z)$ and the observed law is $P_{v,a} = \mathcal{L}(Y)$. We use $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$, equal to half the L^1 distance of densities. Gaussian measure of variance $t \geq 0$ is denoted by γ_t ; $\gamma_0 = \delta_0$.

Lemma 2.1 (Uniform exponential and moment bounds). *For every real t ,*

$$\mathbb{E}e^{tZ} \leq e^{vt^2/2} \leq e^{Vt^2/2}. \quad (2.3)$$

The moment-generating and characteristic functions of Z extend to entire functions. For a constant D depending only on H, V , and every integer $r \geq 1$, $\mathbb{E}|Y|^r \leq (D\sqrt{r})^r$ uniformly over the model.

Proof. The coefficient inequality $(2k+1)! \geq 6^k k!$ follows by induction because $(2k+3)(2k+2) \geq 6(k+1)$. Therefore $\sinh t/t \leq e^{t^2/6}$. Apply this to each finite partial sum and to its independent Gaussian. The partial sums converge in probability, and their exponentials are uniformly integrable: their second exponential moments are bounded by the same inequality at $2t$. Passing to the limit proves (2.3) and the product representation of the MGF.

The bound also holds with t and $-t$, so $\mathbb{E}e^{R|Z|} \leq 2e^{VR^2/2}$ for all R . Differentiation under expectation on compact subsets of $\mathbb{C}$ proves entire extension. Chernoff's inequality gives $\mathbb{P}(|Z| > u) \leq 2e^{-u^2/(2V)}$. Integration of this tail bound yields $\mathbb{E}|Z|^r \leq (D_0\sqrt{r})^r$ after increasing a constant D_0 . The triangle inequality in L^r and $|B| \leq H$ give the assertion for Y . $\square$

Lemma 2.2 (Cumulant coordinates). *Set $\gamma_{2k} = \kappa_{2k}(U_1)$. Then*

$$\gamma_{2k} = \frac{2^{2k} B_{2k}}{2k} = (-1)^{k+1} \frac{2(2k-1)! \zeta(2k)}{\pi^{2k}} \neq 0, \quad (2.4)$$

and

$$\kappa_2(Y) - \kappa_2(B) = v, \quad (2.5)$$

$$\kappa_{2k}(Y) - \kappa_{2k}(B) = \gamma_{2k} p_k(a), \quad k \geq 2. \quad (2.6)$$

The odd latent cumulants vanish. In particular, $\gamma_2 = 1/3$, $\gamma_4 = -2/15$, and $\gamma_6 = 16/63$.

Proof. Expand $\log(\sinh t/t)$ at zero using the classical logarithmic sine expansion [9, Eq. 4.19.7]. Equivalently, Euler's sine product gives $\log(\sinh t/t) = \sum_{k \geq 1} (-1)^{k+1} \zeta(2k) t^{2k} / (k\pi^{2k})$ for $|t| < \pi$. This proves both forms of (2.4). For sufficiently small $|t|$, the sum of these logarithmic series over j converges absolutely, since $\sum a_j^2 < \infty$ and the a_j are bounded. Comparing coefficients of the independent-product MGF proves the formulas. $\square$

Remark 2.3 (The Rvachev background). The random variable $\sum_{j \geq 1} 2^{-j} U_j$ has the normalized Rvachev up density on $[-1, 1]$. Its characteristic function is $\prod_{j \geq 1} \text{sinc}(2^{-j} t)$ in our ordinary characteristic-function convention, and its variance is $1/9$. It is an admissible known background in (2.1), as in the inspected Gaussian-confounding report [3]. Alternatively, the same geometric spectrum can be the *unknown latent* uniform component. These two roles must not be confused.

3 A spectral measure for the Gaussian component

Definition 3.1 (Variance-weighted spectral measure). For the parameters in (2.2), define a finite measure on $[0, L]$ by

$$\rho_{v,a} = 3s \delta_0 + \sum_{j \geq 1} x_j \delta_{x_j}. \quad (3.1)$$

Zero coordinates contribute no mass. At any positive x , the mass is x times the finite integer multiplicity of that squared scale.

Theorem 3.2 (Moment representation and exact identification). *The moments $m_r = \int x^r d\rho_{v,a}(x)$ are*

$$m_0 = 3v, \quad m_r = p_{r+1}(a) \quad (r \geq 1). \quad (3.2)$$

Consequently the exact observed law determines $\rho_{v,a}$, all the positive half-lengths with multiplicity, and $s = \rho_{v,a}(\{0\})/3$.

Proof. The mass identity is $3s + p_1 = 3v$; all positive-degree moments kill the atom at zero and give (3.2). The cumulant formulas recover these moments from the observed law after subtracting the known background cumulants. Two finite measures on $[0, L]$ with the same moments integrate every polynomial equally. Uniform polynomial approximation of continuous functions, followed by uniqueness of finite Borel measures, makes them equal. At each positive atom x , divide its mass by x to recover its multiplicity, and take its positive square root to recover the half-length. The atom at zero gives s . $\square$

This is a moment-theoretic representation of the classical identification result, not a new claim that generalized uniform convolutions are identifiable. Its additional usefulness is that Gaussian variance is now a boundary-atom estimation problem for a positive measure whose entire moment sequence is known in principle.

Proposition 3.3 (Intrinsic Gaussian divisibility). *For every parameter in the model,*

$$s = \lim_{t \rightarrow +\infty} \frac{2 \log \mathbb{E} e^{tY}}{t^2} = \max\{w \geq 0 : P_{v,a} = \gamma_w * Q \text{ for a probability measure } Q\}. \quad (3.3)$$

Proof. For each fixed a_j , $t^{-2} \log(\sinh(a_j t)/(a_j t)) \rightarrow 0$; it is nonnegative and bounded by $a_j^2/6$. Dominated convergence for the sum shows that the uniform component contributes $o(t^2)$ to the log MGF. The bounded background contributes at most $H|t|$ in absolute value. The Gaussian contributes $st^2/2$, proving the limit.

The decomposition with $w = s$ exists by construction. If $P_{v,a} = \gamma_w * Q$, Tonelli's theorem applied to exponential moments shows that Q has finite MGF at every real t and $\log M_Q(t) = \log M_Y(t) - wt^2/2$. Its mean is zero. Jensen's inequality gives $\log M_Q(t) \geq 0$. The displayed limit therefore implies $w \leq s$. $\square$

4 The known compactification and its statistical topology

Define

$$\mathcal{K}_V = \left\{ (v, a) : 0 \leq v \leq V, \quad a_1 \geq a_2 \geq \cdots \geq 0, \quad \sum_{j \geq 1} a_j^2 \leq 3v \right\}, \quad (4.1)$$

with metric $d_\star((v, a), (w, c)) = |v - w| + \sup_j |a_j - c_j|$. Here $s(v, a) = v - p_1(a)/3$. The point of retaining v , rather than s , as a coordinate is that variance carried by disappearing factors is not lost. The following proposition supplies a self-contained smoothed version of the Billey–Swanson framework [6].

Proposition 4.1 (Compact model and a total-variation homeomorphism). *The space $\mathcal{K}_V$ is compact. Its map to latent laws is continuous and injective for weak convergence, and its map to observed laws is a homeomorphism onto a compact subset of the total-variation metric space. Finite uniform sums with $s = 0$ are dense in this observed-law space.*

Proof. Monotonicity gives $a_j^2 \leq L/j$. A diagonal subsequence of any parameter sequence converges in each coordinate and in v ; Fatou's lemma preserves the defining inequality. The uniform tail bound $a_j \leq \sqrt{L/j}$ upgrades coordinate convergence to convergence in the supremum norm. This proves compactness.

For a complex argument z , write the compensated product

$$\phi_{v,a}(z) = e^{-vz^2/2} \prod_{j \geq 1} \left(e^{a_j^2 z^2/6} \operatorname{sinc}(a_j z) \right). \quad (4.2)$$

On every compact set in z , a factor is $1 + O(a_j^4)$ uniformly for large j , and

$$\sum_{j > J} a_j^4 \leq a_{J+1}^2 \sum_{j > J} a_j^2 \leq \frac{L^2}{J+1}. \quad (4.3)$$

Finite products depend continuously on the coordinates, and the tails converge uniformly to one as $J \rightarrow \infty$. Hence the entire characteristic functions converge locally uniformly when parameters converge. Lévy's continuity theorem gives weak continuity. Injectivity is theorem 3.2, even without the background.

If $\mu_n \Rightarrow \mu$, then for every y , $\int b(y-z) d\mu_n(z) \rightarrow \int b(y-z) d\mu(z)$, because $b(y-\cdot)$ is bounded and continuous. These nonnegative densities all have integral one. Scheffé's lemma gives L^1 convergence: indeed, dominated convergence for $\min(f_n, f) \leq f$ and $\|f_n - f\|_1 = 2 - 2 \int \min(f_n, f)$ proves it directly. Thus the observed-law map is TV-continuous. It is injective by theorem 3.2, and a continuous injection from a compact space into a metric space is a homeomorphism onto its image.

For density, retain the first J half-lengths and put $\tau_J = v - \frac{1}{3} \sum_{j \leq J} a_j^2 \geq 0$. Replace this residual variance by n_J uniforms of half-length $\sqrt{3\tau_J/n_J}$, with $n_J \rightarrow \infty$. Choose n_J so that this half-length tends to zero, and sort the combined finite list. Its total variance is v , its Gaussian variance is zero, and each ordered coordinate tends to the target coordinate. The same uniform-tail argument yields $d_\star$ convergence, hence TV convergence of the observed laws. $\square$

Theorem 4.2 (Exactly where Gaussian variance is continuous). *The function $s : \mathcal{K}_V \rightarrow [0, V]$ is upper semicontinuous. Its continuity points are exactly the parameters with $s = 0$. At a parameter with Gaussian variance s , the possible limits of $s(\theta_n)$ along sequences $\theta_n \rightarrow \theta$ are precisely the interval $[0, s]$.*

Proof. The map p_1 is the increasing supremum of continuous finite sums, so it is lower semicontinuous; $s = v - p_1/3$ is upper semicontinuous. At $s = 0$, nonnegativity and upper semicontinuity imply continuity.

Fix a target and a number $u \in [0, s]$. Keep the original uniform spectrum, replace Gaussian variance $s - u$ by n equal uniform factors of half-length $\sqrt{3(s-u)/n}$, and keep Gaussian variance u . Sorting changes no law. The total variance stays v , and the ordered coordinates converge to the original spectrum. The Gaussian variances are constantly u . This proves attainability of every proposed limit and discontinuity when $s > 0$. Upper semicontinuity and nonnegativity exclude all other limits. $\square$

Remark 4.3. The full parameter (v, a) is uniformly recoverable as a *continuous function of the exact law* in the compactified metric. The functional s is not continuous on that parameter space. There is no contradiction: infinitely many small coordinate changes may carry a fixed sum of squares. At a zero-Gaussian target, convergence also holds in ℓ^2 for the spectra, since their squared norms converge and their coordinates converge.

5 A rank-free expansion for Gaussian dust

The compactification describes convergence but does not give a numerical error. The next theorem quantifies how a cloud of small uniform factors replaces a Gaussian component. The background in this section need only be a probability density $b \in W^{8,1}(\mathbb{R})$; compact support is not required. Write $\|b^{(r)}\|_1$ for the L^1 norm of its weak derivative.

Theorem 5.1 (A uniform fourth-order smoothing expansion). *Let a be any finite or square-summable sequence of nonnegative half-lengths, and put*

$$f = b * \mathcal{L} \left(\sum_j a_j U_j \right), \quad g = b * \gamma_{p_1/3}.$$

Then

$$\left\| f - g + \frac{p_2}{180} g^{(4)} \right\|_1 \leq \frac{43}{22680} p_3 \|b^{(6)}\|_1 + \frac{p_2^2}{4050} \|b^{(8)}\|_1. \quad (5.1)$$

The constants are independent of the number of factors and of their collisions. They are not asserted optimal.

Proof. Let T_i be convolution by $a_i U_i$, let G_i be convolution by $\gamma_{a_i^2/3}$, and set $A_i = T_i - G_i$. All these operators commute with weak derivatives, and T_i, G_i contract L^1 . Taylor's formula for translation, with integral remainder, gives

$$\left\| h(\cdot - z) - \sum_{r=0}^{k-1} \frac{(-z)^r}{r!} h^{(r)} \right\|_1 \leq \frac{|z|^k}{k!} \|h^{(k)}\|_1. \quad (5.2)$$

It applies to Sobolev functions by approximation from smooth functions. The uniform and its variance-matched Gaussian agree in moments through order three. Their fourth moments are $a_i^4/5$ and $a_i^4/3$, respectively. Consequently

$$\|A_i h\|_1 \leq \frac{a_i^4}{45} \|h^{(4)}\|_1, \quad (5.3)$$

$$\left\| A_i h + \frac{a_i^4}{180} h^{(4)} \right\|_1 \leq \frac{11a_i^6}{11340} \|h^{(6)}\|_1. \quad (5.4)$$

For the first line use $(1/5 + 1/3)/4! = 1/45$. For the second line expand through degree five, use the fourth-moment difference $(1/5 - 1/3)/4! = -1/180$, and bound the sixth-moment remainder by $(1/7 + 5/9)/6! = 11/11340$.

First suppose there are n factors. An exact telescoping identity is

$$\prod_{i=1}^n T_i - \prod_{i=1}^n G_i = \sum_{i=1}^n A_i \left(\prod_{j<i} T_j \right) \left(\prod_{j>i} G_j \right).$$

Replace the product $\prod_{j<i} T_j$ by $\prod_{j<i} G_j$ with a second telescoping identity. Every resulting remainder contains $A_i A_j$ for one pair $j < i$, and all other factors are L^1 contractions. Applying (5.3) twice bounds its action on b by $a_i^4 a_j^4 \|b^{(8)}\|_1 / 45^2$. Thus

$$f - g = \sum_i A_i \left(\prod_{j \neq i} G_j \right) b + R, \quad \|R\|_1 \leq \frac{p_2^2}{4050} \|b^{(8)}\|_1. \quad (5.5)$$

There are no uncontrolled higher-order subset sums in this argument.

Apply (5.4) to each summand. To replace $\left(\prod_{j \neq i} G_j\right) b^{(4)}$ by $g^{(4)}$, use centering and (5.2) at order two:

$$\|(I - G_i)h\|_1 \leq \frac{a_i^2}{6} \|h''\|_1.$$

Its contribution, after multiplication by $a_i^4/180$, is at most $a_i^6 \|b^{(6)}\|_1 / 1080$. Since

$$\frac{11}{11340} + \frac{1}{1080} = \frac{43}{22680},$$

we obtain (5.1) for a finite list.

For an infinite list, the uniform partial sums converge weakly, and convolution with any L^1 function is continuous from weak convergence into L^1 . For completeness, approximate that function in L^1 by a continuous compactly supported function; for the latter, tightness controls spatial tails and dominated convergence applies on a compact interval. Convolution contractions control the approximation errors. Apply this observation to b and its derivatives. Gaussian convolutions have the same continuity, and the finite p_1, p_2, p_3 sums converge to their infinite counterparts. Pass to the limit in the finite-list inequality. $\square$

Corollary 5.2 (Sharp first response of a small-factor cloud). *For a sequence of spectra with $h = \sup_j a_j \rightarrow 0$, $p_1 \rightarrow 3\tau < \infty$, and $p_2 > 0$, one has*

$$\text{TV} \left(b * \mathcal{L} \left(\sum_j a_j U_j \right), b * \gamma_{p_1/3} \right) \sim \frac{p_2}{360} \left\| (b * \gamma_\tau)^{(4)} \right\|_1. \quad (5.6)$$

The constant multiplying p_2 is strictly positive.

Proof. The inequalities $p_3 \leq h^2 p_2$ and $p_2 \leq h^2 p_1$ make the remainder in (5.1) equal to $o(p_2)$. Moreover, $(b * \gamma_{p_1/3})^{(4)} \rightarrow (b * \gamma_\tau)^{(4)}$ in L^1 . If the latter derivative were zero, the density would be a polynomial of degree at most three in the distributional sense. An integrable polynomial on $\mathbb{R}$ must be zero, contradicting mass one. Divide the expansion by p_2 , take L^1 norms, and divide by two. $\square$

Equal dust. Take n factors with $a_j = \sqrt{3\delta/n}$. They have variance δ , $p_2 = 9\delta^2/n$, and $p_3 = 27\delta^3/n^2$. Therefore

$$\text{TV} \left(b * \mathcal{L} \left(\sqrt{3\delta/n} \sum_{j=1}^n U_j \right), b * \gamma_\delta \right) \sim \frac{\delta^2}{40n} \left\| (b * \gamma_\delta)^{(4)} \right\|_1. \quad (5.7)$$

A fixed collection of macroscopic factors, or a fixed additional Gaussian, may be absorbed into b ; its derivative norms can only decrease under such convolution. This gives a quantitative version of the perturbations used in theorem 4.2.

6 Exact finite-sample impossibility without a tail condition

Let $Y_1, \dots, Y_N$ be independent with common law P_θ , $\theta = (v, a) \in \mathcal{K}_V$. Estimators may be any measurable functions of the data; randomization does not improve the lower bounds below.

Theorem 6.1 (No uniform statistical gain at any finite sample size). *For every integer $N \geq 1$,*

$$\inf_{\hat{s}} \sup_{\theta \in \mathcal{K}_V} \mathbb{E}_{\theta} |\hat{s} - s(\theta)| = \frac{V}{2}, \quad (6.1)$$

$$\inf_{\hat{s}} \sup_{\theta \in \mathcal{K}_V} \mathbb{E}_{\theta} (\hat{s} - s(\theta))^2 = \frac{V^2}{4}. \quad (6.2)$$

Both upper bounds are attained by the constant estimator $\hat{s} = V/2$. The lower bounds hold even on the fixed-total-variance slice $v = V$.

Proof. Compare the parameter θ_G with no uniform factors and Gaussian variance V to θ_n with zero Gaussian variance and n equal half-lengths $\sqrt{3V/n}$. Both have total variance V . By theorem 4.1, or by (5.7), $\text{TV}(P_{\theta_n}, P_{\theta_G}) \rightarrow 0$. A product coupling, or telescoping product densities, gives

$$\text{TV}(P_{\theta_n}^{\otimes N}, P_{\theta_G}^{\otimes N}) \leq N \text{TV}(P_{\theta_n}, P_{\theta_G}) \rightarrow 0$$

for every fixed N .

Clipping an estimator into $[0, V]$ cannot increase either loss. Let f_n, f_G be the two sample densities, and let z be the clipped estimate at a given sample. The sum of the absolute-error risks is at least

$$\int \min(f_n, f_G)(|z| + |z - V|) = V(1 - \text{TV}(P_{\theta_n}^{\otimes N}, P_{\theta_G}^{\otimes N})).$$

The larger risk is at least half this quantity. Sending $n \rightarrow \infty$ gives $V/2$. For squared error, use $z^2 + (z - V)^2 \geq V^2/2$ in the same calculation to obtain $V^2/4$. The constant estimator gives the matching upper bounds. $\square$

Corollary 6.2 (The corresponding testing obstruction). *In testing the single law with $s = V, a = 0$ against all laws with $s = 0, v = V$, the infimum, over tests, of type-I error plus worst-case type-II error equals one at every finite sample size.*

Proof. The sum of the errors for a simple pair is at least one minus their TV distance. Use the preceding sequence of alternatives. A constant test attains one. $\square$

Remark 6.3 (What this does and does not say). Exact identification from an entire distribution is compatible with theorem 6.1. The worst parameter is allowed to depend on the sample size, and can contain as many tiny factors as needed. This is not a claim that every fixed parameter is statistically indistinguishable from every other one, or that pointwise consistent estimation is impossible. It also does not contradict the repository's fixed-capacity rates: the required number of factors here has no a priori bound.

Corollary 6.4 (Local minimax risk at an individual law). *Fix $\theta \in \mathcal{K}_V$ with Gaussian variance s_0 . For $\eta > 0$, restrict the minimax problem to parameters θ' with $\text{TV}(P_{\theta'}, P_{\theta}) < \eta$. For each fixed sample size N , as $\eta \downarrow 0$ the absolute-error minimax risk tends to $s_0/2$, and the squared-error minimax risk tends to $s_0^2/4$.*

Proof. Replace all Gaussian variance s_0 at θ by equal dust, keeping its original uniforms. This produces parameters of Gaussian variance zero whose observed laws converge in TV to P_{θ} . Eventually they lie in any fixed neighborhood, and the two-point proof of theorem 6.1 yields lower bounds $s_0/2$ and $s_0^2/4$. Upper semicontinuity of s , transferred to the observed-law space by theorem 4.1, implies that for every $\varepsilon > 0$ all sufficiently small neighborhoods have $0 \leq s(\theta') \leq s_0 + \varepsilon$. The appropriate midpoint constant estimator gives upper bounds $(s_0 + \varepsilon)/2$ and $(s_0 + \varepsilon)^2/4$. Send ε to zero. $\square$

7 Positive finite-cumulant certificates

A polynomial that equals one at zero but suppresses positive spectral atoms provides an upper bound on the Gaussian variance. The simplest such family already resolves the qualitative recovery question.

Theorem 7.1 (A monotone cumulant ladder). *For $m \geq 0$, define*

$$A_m = v - \frac{1}{3} \sum_{r=1}^m (-1)^{r+1} \binom{m}{r} \frac{p_{r+1}}{L^r}, \quad A_0 = v. \quad (7.1)$$

These quantities depend on cumulants only through order $2m + 2$. They satisfy

$$A_m - s = \frac{1}{3} \sum_{j \geq 1} x_j (1 - x_j/L)^m \geq 0, \quad A_m \downarrow s. \quad (7.2)$$

For every $J, m \geq 1$,

$$3(A_m - s) \leq \frac{JL}{em} + \sum_{j > J} x_j. \quad (7.3)$$

Conversely,

$$\sum_{j > 2m} x_j \leq 6(A_m - s), \quad m \geq 1. \quad (7.4)$$

Proof. Integrate $(1 - x/L)^m$ against (3.1), expand by the binomial formula, and divide by three. The atom at zero contributes s . For each positive x_j , $(1 - x_j/L)^m \downarrow 0$, and the summable majorant x_j permits dominated convergence. For (7.3), bound the first J terms by $xe^{-mx/L} \leq L/(em)$ each, and keep the remaining terms unchanged. For $j > 2m$, $x_j \leq L/j \leq L/(2m + 1)$, so Bernoulli's inequality gives $(1 - x_j/L)^m \geq 1 - m/(2m + 1) > 1/2$. Substitute in (7.2). $\square$

Every A_m is continuous on $\mathcal{K}_V$. Indeed v is continuous, and for $k \geq 2$ the tail of p_k is bounded by $L^k/(J + 1)^{k-1}$, uniformly over $\mathcal{K}_V$. The unknown p_1 never appears in (7.1); it would be circular to use it as input to an estimator of s .

8 A necessary and sufficient condition for uniform recovery

Theorem 8.1 (Complete compact-class criterion). *Let K be a nonempty compact subset of $\mathcal{K}_V$, in the metric $d_\star$. The following statements are equivalent:*

- (i) *The Gaussian variance s is continuous on K .*
- (ii) *The uniform-factor energy tails vanish uniformly:*

$$\lim_{J \rightarrow \infty} \sup_{(v, a) \in K} \sum_{j > J} a_j^2 = 0. \quad (8.1)$$

- (iii) *The finite-cumulant ladder A_m converges uniformly to s on K .*
- (iv) *There exist estimators from N independent observations with*

$$\sup_{\theta \in K} \mathbb{E}_\theta |\hat{s}_N - s(\theta)| \longrightarrow 0. \quad (8.2)$$

Proof. For (i) $\Rightarrow$ (ii), $p_1 = 3(v - s)$ is continuous on K and its finite partial sums are continuous and increase pointwise to p_1 . Dini's theorem gives uniform convergence. For (ii) $\Rightarrow$ (i), the uniform limit of these continuous partial sums is continuous, hence so is s .

The implication (ii) $\Rightarrow$ (iii) follows by first choosing J in (7.3) and then m . Conversely (7.4) proves (iii) $\Rightarrow$ (ii). Alternatively, (i) $\Rightarrow$ (iii) follows from Dini's theorem applied to $A_m \downarrow s$.

For (iii) $\Rightarrow$ (iv), fix m . Estimate the finitely many raw moments up to order $2m + 2$ by sample averages and clip them to fixed compact intervals containing all the corresponding true moments on $\mathcal{K}_V$. Uniform moment bounds from theorem 2.1 make the expected moment errors $O_m(N^{-1/2})$. Cumulants are polynomials in these moments, so their errors have the same bound after clipping; therefore the estimator of A_m obtained from (7.1) has uniformly vanishing expected error. Choose increasing integers N_m so large that this error is at most $1/m$ for all $N \geq N_m$. Let $m(N)$ increase sufficiently slowly along these thresholds. Uniform convergence of $A_m - s$ and clipping the final answer into $[0, V]$ prove (8.2).

Finally assume (iv), and clip the estimators into $[0, V]$. For each fixed N , $\theta \mapsto \mathbb{E}_\theta \hat{s}_N$ is continuous on K : theorem 4.1 gives TV continuity of the one-observation laws, tensorization gives it for their N -fold products, and expectations of a bounded statistic are TV-continuous. By (iv), these continuous functions converge uniformly to s . Hence s is continuous. $\square$

Corollary 8.2 (Useful sufficient classes). *Uniform consistency holds on every compact subclass with any one of the following bounds, with constants independent of the parameter: a fixed finite capacity; $a_j \leq Cj^{-\beta}$ for $\beta > 1/2$; $a_j \leq Cq^j$ for $0 < q < 1$; or $\sum_j a_j \leq A$.*

Proof. The first three bounds give a summable deterministic majorant for a_j^2 . For the last, monotonicity gives $a_j \leq A/j$, so $\sum_{j>J} a_j^2 \leq a_{J+1} \sum_{j>J} a_j \leq A^2/(J+1)$. Apply theorem 8.1. $\square$

Remark 8.3 (Compactness and quantifiers matter). An arbitrary nonclosed parameter class need not satisfy this equivalence: continuity at each of its points does not imply a uniform modulus. The theorem requires compactness in the specified topology, not merely a bounded list of individual variances. The full $\mathcal{K}_V$ violates the tail condition: for any fixed J , a sufficiently long equal-factor cloud can place almost all of its energy beyond J .

9 Christoffel recovery from higher cumulants

The binomial ladder is explicit, but it uses a prescribed polynomial. Optimizing the polynomial yields the classical Christoffel function at zero, applied here to the spectral measure rather than to the observed density. This distinction is essential: empirical moments of Y are not the moments of ρ until they have been converted by theorem 2.2.

Definition 9.1 (Gaussian-variance Christoffel certificate). For an integer $d \geq 0$, put

$$\mathcal{C}_d(v, a) = \frac{1}{3} \inf \left\{ \int q(x)^2 d\rho_{v,a}(x) : q \in \mathbb{R}[x], \deg q \leq d, q(0) = 1 \right\}. \quad (9.1)$$

Theorem 9.2 (Hierarchy, matrix formula, and tail certificate). *The quantities $\mathcal{C}_d$ depend on observed cumulants only through order $4d + 2$. They decrease to s and satisfy*

$$0 \leq \mathcal{C}_d - s \leq \frac{1}{3} \sum_{j>d} x_j. \quad (9.2)$$

For $d \geq 1$, define

$$H_d = (p_{i+j+1})_{i,j=1}^d, \quad y_d = (p_{i+1})_{i=1}^d. \quad (9.3)$$

Then y_d belongs to the range of the positive semidefinite matrix H_d , and

$$\mathcal{C}_d = v - \frac{1}{3} y_d^\top H_d^\dagger y_d, \quad (9.4)$$

where $H_d^\dagger$ is the Moore–Penrose inverse; $\mathcal{C}_0 = v$. Moreover $\mathcal{C}_d = s$ if and only if the spectrum has at most d distinct positive squared scales.

Proof. Write $\mu_a = \sum_j x_j \delta_{x_j}$. Since $q(0) = 1$,

$$\mathcal{C}_d = s + \frac{1}{3} \lambda_d, \quad \lambda_d = \inf_{\deg q \leq d, q(0)=1} \int q^2 d\mu_a. \quad (9.5)$$

For $q(x) = 1 + \sum_{i=1}^d c_i x^i$, the integral against ρ is

$$3v + 2c^\top y_d + c^\top H_d c. \quad (9.6)$$

This uses moments $m_0, \dots, m_{2d}$ and hence $v, p_2, \dots, p_{2d+1}$, not the unknown p_1 . If $u \in \ker H_d$, the polynomial $\sum_i u_i x^i$ has zero squared integral against μ_a , so it vanishes μ_a -almost everywhere. Its integral against μ_a is therefore $u^\top y_d = 0$. Thus y_d is orthogonal to $\ker H_d$ and lies in its range. Completing the square on that range shows that the minimum is attained and proves (9.4).

If $x_1, \dots, x_d$ are positive, use the test polynomial

$$q_d(x) = \prod_{i=1}^d (1 - x/x_i). \quad (9.7)$$

It vanishes at the first d entries, and for every remaining x_j each factor lies in $[0, 1]$. Its squared integral is at most $\sum_{j>d} x_j$. If there are fewer positive entries, use their finite product instead. This proves (9.2); summability makes its right-hand side tend to zero. Monotonicity follows by increasing the admissible polynomial space.

If there are at most d distinct positive squared scales, a polynomial vanishing at each of them gives $\lambda_d = 0$. Conversely, the attained minimum can be zero only if an admissible polynomial vanishes at every positive atom. It has at most d distinct real roots. This proves the final assertion, including $d = 0$. $\square$

Remark 9.3 (A small-order formula). Provided $p_3 > 0$,

$$\mathcal{C}_1 = v - \frac{p_2^2}{3p_3}. \quad (9.8)$$

It is exact for any number of equal positive half-lengths, because only one distinct squared scale is present. At the zero spectrum it is interpreted by the pseudoinverse formula and equals $v = s$. The numerator and denominator are recovered from the fourth and sixth cumulants. This formula should not be implemented as an unregularized ratio when p_3 is small.

Remark 9.4 (What the optimization relaxes). The Christoffel problem is a positive-measure polynomial problem. It does not impose the extra integer-multiplicity restriction $\rho(\{x\})/x \in \mathbb{N}$ at positive atoms. Therefore its upper certificate is not asserted to be the largest Gaussian variance among all uniform-factor models matching a finite list of cumulants. That sharper constrained inverse problem remains separate.

10 An exact geometric formula

Suppose the unknown squared scales form an exact geometric sequence,

$$x_j = Cr^j, \quad j = 0, 1, 2, \dots, \quad C > 0, \quad 0 < r < 1. \quad (10.1)$$

The reindexing from $j = 1$ elsewhere is only for convenience. Here $p_k = C^k/(1 - r^k)$ and the uniform variance is $C/[3(1 - r)]$.

Theorem 10.1 (Exact finite-degree Christoffel bias). *For the spectrum (10.1), at every integer $d \geq 0$,*

$$\boxed{\mathcal{C}_d - s = \frac{C(1 - r)r^d}{3(1 - r^{d+1})^2}}. \quad (10.2)$$

In particular, $\mathcal{C}_d - s \sim C(1 - r)r^d/3$ as $d \rightarrow \infty$.

Proof. All finite moment matrices of μ_a are positive definite because its support is infinite. Completing the square in the constant coefficient gives

$$\lambda_d = \frac{\det (C^{i+j+1}/(1 - r^{i+j+1}))_{i,j=0}^d}{\det (C^{i+j+1}/(1 - r^{i+j+1}))_{i,j=1}^d}, \quad (10.3)$$

with the empty denominator determinant equal to one.

We use the elementary Cauchy identity

$$\det \left(\frac{1}{1 - u_i v_j} \right)_{i,j=0}^d = \frac{\prod_{0 \leq i < k \leq d} (u_i - u_k) \prod_{0 \leq j < \ell \leq d} (v_j - v_\ell)}{\prod_{i,j=0}^d (1 - u_i v_j)}. \quad (10.4)$$

One proof applies the ordinary Cauchy determinant to $1/(u_i^{-1} - v_j)$ and divides row i by u_i ; the powers of the u_i cancel. The ordinary identity follows by multiplying the determinant by its common denominator: the numerator is alternating in each variable list, hence divisible by both Vandermonde products; comparison at a simple pole, inductively on the matrix size, gives coefficient one. This argument proves (10.4) for the nonzero distinct variables used here, with all denominators nonzero.

In (10.3), row and column factors of C leave a single factor C in the quotient. Apply (10.4) with $u_i = r^i$ and $v_j = r^{j+1}$. The factors introduced by row and column zero in the numerator are

$$\prod_{i=1}^d (1 - r^i) \prod_{j=1}^d (r - r^{j+1}) = r^d \prod_{i=1}^d (1 - r^i)^2.$$

The introduced denominator factors are

$$\prod_{j=0}^d (1 - r^{j+1}) \prod_{i=1}^d (1 - r^{i+1}) = (1 - r) \prod_{i=2}^{d+1} (1 - r^i)^2.$$

Their quotient is $r^d(1 - r)/(1 - r^{d+1})^2$. Multiply by C and use (9.5) to prove the theorem. $\square$

10.1 The dyadic Fabius–Rvachev specialization

For half-lengths $a_j = 2^{-j}$, $j \geq 1$, set $C = r = 1/4$. The exact result is

$$\mathcal{C}_d - s = \frac{4^{-d}}{16(1 - 4^{-d-1})^2}. \quad (10.5)$$

The first few values are particularly simple:

d	Highest cumulant order	Exact bias	Decimal bias
0	2	$1/9$	0.1111111111
1	6	$4/225$	0.0177777778
2	10	$16/3969$	0.0040312421
3	14	$64/65025$	0.0009842368
4	18	$256/1046529$	0.0002446182

The accompanying program regenerates these rational values and the decimal roundings. Unlike a finite truncation of the uniform sum, these numbers are exact for the infinite geometric spectrum.

10.2 Interpretation and its limitations

In the zero-based indexing of (10.1), deleting the first d entries leaves the tail $\sum_{j=d}^{\infty} Cr^j = Cr^d/(1-r)$. Consequently (10.2) improves the generic upper bound in (9.2) by the exact factor

$$\frac{\mathcal{C}_d - s}{Cr^d/[3(1-r)]} = \frac{(1-r)^2}{(1-r^{d+1})^2} \rightarrow (1-r)^2. \quad (10.6)$$

This exact formula is not a statistical minimax rate. It describes noiseless finite-moment bias. If exact geometric shape with unknown C, r were imposed as a parametric assumption, those few parameters could instead be recovered from a few cumulants. The general-purpose certificate deliberately does not use that parametric shortcut.

11 Regularization by bounded polynomial coefficients

Direct inversion of H_d is unnecessary for a robust upper certificate. Normalize spectral locations by $z = x/L$ and define

$$\bar{m}_0 = 3v, \quad \bar{m}_r = p_{r+1}/L^r \quad (r \geq 1). \quad (11.1)$$

These are the moments of the pushforward of ρ to $[0, 1]$; the total mass is still $3v$, not one. For $M \geq 1$, let

$$\mathcal{Q}_{d,M} = \left\{ q(z) = \sum_{i=0}^d c_i z^i : c_0 = 1, \sum_{i=0}^d |c_i| \leq M \right\}. \quad (11.2)$$

This is a compact coefficient set. Given estimates $\hat{m}_0, \dots, \hat{m}_{2d}$, define

$$\hat{C}_{d,M} = \frac{1}{3} \inf_{q \in \mathcal{Q}_{d,M}} \sum_{i,j=0}^d c_i c_j \hat{m}_{i+j}. \quad (11.3)$$

The value is a continuous function of the estimated moments, even if their Hankel matrix is not positive semidefinite. The estimated quadratic may then be nonconvex; existence of its global minimum still follows from compactness. No efficient general nonconvex solver is implicit in this definition.

Theorem 11.1 (A deterministic noise certificate). *Suppose*

$$\max_{0 \leq r \leq 2d} |\hat{m}_r - \bar{m}_r| \leq \varepsilon. \quad (11.4)$$

Let $C_{d,M}$ be (11.3) with the true moments. Then

$$|\hat{C}_{d,M} - C_{d,M}| \leq \frac{\varepsilon M^2}{3}. \quad (11.5)$$

If $x_i \geq \ell_i > 0$ for $1 \leq i \leq d$ and

$$M \geq M_d := \prod_{i=1}^d (1 + L/\ell_i), \quad (11.6)$$

then

$$|\text{clip}_{[0,V]}(\hat{C}_{d,M}) - s| \leq \frac{1}{3} \sum_{j>d} x_j + \frac{\varepsilon M^2}{3}. \quad (11.7)$$

Moreover

$$U_{d,M} := \text{clip}_{[0,V]} \left(\hat{C}_{d,M} + \frac{\varepsilon M^2}{3} \right) \quad (11.8)$$

is an upper bound on s , and $0 \leq U_{d,M} - s \leq \frac{1}{3} \sum_{j>d} x_j + 2\varepsilon M^2/3$.

Proof. For any $q \in \mathcal{Q}_{d,M}$, the difference between the estimated and true quadratic is at most $\varepsilon \sum_{i,j} |c_i| |c_j| \leq \varepsilon M^2$. The infima differ by at most the same amount, proving (11.5). The polynomial $\prod_{i=1}^d (1 - Lz/x_i)$ has coefficient ℓ^1 norm exactly $\prod_{i=1}^d (1 + L/x_i) \leq M_d$, because its coefficients alternate in sign. It is therefore admissible. The argument of theorem 9.2 gives $0 \leq C_{d,M} - s \leq \frac{1}{3} \sum_{j>d} x_j$. Combine this with (11.5). Clipping into an interval containing s does not increase absolute error, and preserves the certified upper-bound property. $\square$

Remark 11.2 (A fully specified, though possibly expensive, computation). One may approximate the infimum in (11.3) over a finite rational grid in the coefficient ball. If the coefficient ℓ^1 mesh is η , and $K_0 = \max_r |\hat{m}_r|$, the objective changes by at most $2K_0 M \eta$ between two grid-neighbor coefficient vectors, since $|c^\top \hat{H} c - d^\top \hat{H} d| \leq K_0 \|c - d\|_1 (\|c\|_1 + \|d\|_1)$. A grid fine enough to meet a specified objective tolerance therefore gives a finite globally certified procedure. Such a search is not promised to be practical in high degree; the theorem is an error guarantee, not a complexity claim.

12 Statistical upper bounds on envelope classes

We now turn the deterministic certificate into actual estimation guarantees. These bounds do not claim optimal rates. Their purpose is to prove a constructive contrast with theorem 6.1 and quantify the price of infinite rank under explicit assumptions.

Lemma 12.1 (Uniform estimation of the normalized spectral moments). *For each $d \geq 1$, there are estimates of $\bar{m}_0, \dots, \bar{m}_{2d}$ from N observations such that*

$$\sup_{\theta \in \mathcal{K}_V} \mathbb{E}_\theta \max_{0 \leq r \leq 2d} |\hat{m}_r - \bar{m}_r| \leq N^{-1/2} \exp\{C_0 d \log(d+1)\}, \quad (12.1)$$

where C_0 depends only on H, V and the known background, not on d, N .

Proof. Set $R = 4d + 2$. Increase a constant D_1 so that, with $M_R = D_1\sqrt{R} \geq 1$, every raw moment $\mu_j = \mathbb{E}Y^j$, $j \leq R$, has $|\mu_j| \leq M_R^j$ and $\mathbb{E}|Y|^{2j} \leq M_R^{2j}$. This follows from theorem 2.1, using $2j \leq 2R$ and absorbing $\sqrt{2}$ in D_1 . Estimate μ_j by the sample mean of Y^j and clip it into $[-M_R^j, M_R^j]$. The resulting $\hat{\mu}_j$ satisfies

$$\mathbb{E}|\hat{\mu}_j - \mu_j| \leq M_R^j/\sqrt{N}. \quad (12.2)$$

For $r \leq R$, use the standard partition expression for a cumulant,

$$\kappa_r = \sum_{\pi \in \Pi_r} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{A \in \pi} \mu_{|A|}, \quad (12.3)$$

which follows by comparing coefficients in $\log(\sum_{j \geq 0} \mu_j t^j / j!)$. A telescoping product and (12.2) bound the expected error of the product for a partition π by $|\pi| M_R^r / \sqrt{N}$. Furthermore,

$$\sum_{\pi \in \Pi_r} (|\pi| - 1)! |\pi| = \sum_{q=1}^r S(r, q) q! \leq \sum_{q=1}^r q^r \leq r^{r+1},$$

because $S(r, q)q!$ counts surjections to q labeled points. Thus the expected cumulant error is at most $r^{r+1} M_R^r / \sqrt{N}$.

Subtract the exact known cumulants of B . Use (2.5) for $\bar{m}_0 = 3v$ and (2.6) for $\bar{m}_r$, $r \geq 1$. By (2.4), $|\gamma_{2k}|^{-1} \leq \pi^{2k}/2$. The factors L^{-r} and these reciprocal cumulants grow at most exponentially in R , since $V > 0$ is fixed. Sum the expected errors to bound the expected maximum. All factors are absorbed in $\exp\{C_0 d \log(d+1)\}$, proving the lemma. $\square$

Theorem 12.2 (Constructive risk bound). *Suppose a class satisfies $x_i \geq \ell_i > 0$ for $i \leq d$ and $\sum_{j>d} x_j \leq T_d$ uniformly. Using the estimates in theorem 12.1, the clipped estimator (11.3) with $M = M_d$ satisfies*

$$\sup_{\theta} \mathbb{E}_{\theta} |\text{clip}_{[0,V]}(\hat{C}_{d,M_d}) - s| \leq \frac{T_d}{3} + \frac{M_d^2}{3\sqrt{N}} \exp\{C_0 d \log(d+1)\}. \quad (12.4)$$

A finite coefficient-grid minimization with objective tolerance N^{-1} adds at most $1/(3N)$ to this bound.

Proof. The deterministic argument of theorem 11.1 holds for every realized moment error, not only on a fixed high-probability event. Take expectations and apply theorem 12.1. The grid tolerance statement follows directly from division of the objective value by three. $\square$

Corollary 12.3 (Polynomial envelopes). *Suppose the sorted half-lengths satisfy*

$$c_- j^{-\beta} \leq a_j \leq c_+ j^{-\beta} \quad (j \geq 1), \quad c_- > 0, \quad \beta > 1/2, \quad (12.5)$$

with fixed constants, and the class is nonempty under the variance budget. There exist estimators for which, as $N \rightarrow \infty$,

$$\sup_{\theta} \mathbb{E}_{\theta} |\hat{s}_N - s| = O\left(\left(\frac{\log N}{\log \log N}\right)^{1-2\beta}\right). \quad (12.6)$$

Proof. Take $\ell_i = c_-^2 i^{-2\beta}$. Then

$$\log M_d \leq d \log(1 + L/c_-^2) + 2\beta \log(d!) = O(d \log(d+1)), \quad T_d \leq \frac{c_+^2}{2\beta - 1} d^{1-2\beta}.$$

Choose $d = \lfloor c \log N / \log \log N \rfloor$ with $c > 0$ sufficiently small relative to the constants in (12.4). Its noise term is $O(N^{-1/4})$ for all sufficiently large N ; its bias term is the displayed bound. Handle finitely many small N by the constant estimator. $\square$

Corollary 12.4 (Geometric envelopes). *Suppose*

$$c_- q^j \leq a_j \leq c_+ q^j \quad (j \geq 1), \quad c_- > 0, \quad 0 < q < 1, \quad (12.7)$$

with fixed constants and nonempty model class. There are constants $c, C > 0$ and estimators such that

$$\sup_{\theta} \mathbb{E}_{\theta} |\hat{s}_N - s| \leq C \exp\{-c\sqrt{\log N}\} \quad (12.8)$$

for all sufficiently large N .

Proof. Take $\ell_i = c_-^2 q^{2i}$. Then

$$\log M_d \leq d \log(1 + L/c_-^2) + \log(1/q)d(d+1) = O(d^2), \quad T_d \leq \frac{c_+^2 q^{2d+2}}{1 - q^2}.$$

Set $d = \lfloor c_1 \sqrt{\log N} \rfloor$ with c_1 sufficiently small. The noise term in (12.4) is $O(N^{-1/4})$ for large N , and the tail term is bounded by the right-hand side of (12.8). $\square$

Remark 12.5 (Precisely what assumptions these rates use). The positive lower envelopes control the coefficients of the annihilating polynomial. They are not needed for the qualitative uniform-consistency criterion. Neither rate is asserted minimax, and neither is a rate for estimating the entire ordered spectrum. These classes allow independent deviations from geometric or polynomial shape at every coordinate; they are not finite-dimensional parametric families. The constants depend on the fixed envelope and may deteriorate badly as its lower bound decreases.

13 Pointwise estimation despite constant global minimax risk

The global obstruction concerns a supremum over parameters; it should not be mistaken for an impossibility of learning any single fixed distribution. The positive ladder produces one sequence of estimators that is consistent at every fixed member of the full unrestricted class.

Theorem 13.1 (One pointwise consistent estimator on the whole class). *Let $m = m(N) = \lfloor \sqrt{\log(N+e)} \rfloor$. Use theorem 12.1 with $d = m$ to estimate the normalized moments, and put*

$$\hat{s}_N = \text{clip}_{[0,V]} \left(\frac{1}{3} \sum_{r=0}^m (-1)^r \binom{m}{r} \hat{m}_r \right). \quad (13.1)$$

For every fixed $\theta \in \mathcal{K}_V$, $\mathbb{E}_{\theta} |\hat{s}_N - s(\theta)| \rightarrow 0$. More precisely, uniformly over θ ,

$$\mathbb{E}_{\theta} |\hat{s}_N - s(\theta)| \leq A_m(\theta) - s(\theta) + \frac{2^m}{3\sqrt{N}} \exp\{C_0 m \log(m+1)\}. \quad (13.2)$$

Only the first term need fail to converge uniformly.

Proof. The true moment polynomial in (13.1) is A_m . The sum of the absolute binomial coefficients is 2^m , so theorem 12.1 bounds the expected error of this polynomial by the second term of (13.2). Clipping cannot increase the error against s . The second term tends to zero since $m \log(m+1) = o(\log N)$. For each fixed parameter, theorem 7.1 makes the first term tend to zero as well. $\square$

The observations therefore support the following three simultaneous facts: exact-law identifiability; pointwise consistent estimation from increasing samples; and a nonzero, sample-size-independent global minimax risk. The resolution is the absence of a uniform energy-tail modulus. The dust alternative in a worst-case lower bound becomes increasingly complicated as the sample size grows.

14 Verification, reproducibility, and formalization route

14.1 What the accompanying program checks

The file `verify_results.py` uses exact rational arithmetic for algebraic checks and high-precision arithmetic for a separately labeled Fourier diagnostic. It checks the Bernoulli cumulants, all constants in the Taylor remainder, the geometric Schur-complement formula over several rational values of C, r , finite-spectrum pseudoinverse formulas and exact termination, positive-ladder identities, and representative coefficient-noise bounds. It also regenerates the dyadic table. A high-precision calculation of equal-dust characteristic functions tests the sign and coefficient of the first correction. That diagnostic is *not* a numerical computation of total variation.

The delivered run passed 4,619 exact assertions and three high-precision Fourier diagnostic checks. All data written by the script are consequences or diagnostics of the stated formulas. They are not used to bridge an unproved infinite range. In particular, the statistical assertions are justified by the proofs, not by Monte Carlo simulations. The program is an independently executable companion, not a formal proof certificate accepted by Lean.

14.2 A realistic formalization decomposition

A future Lean or Rocq development can separate five layers. First formalize the finite algebraic layer: moment-cumulant conversion, Hankel Gram matrices, a quadratic minimum on a matrix range, and the Cauchy determinant quotient. This would certify the geometric formula and finite spectral computations without any probability theory.

Second formalize square-summable independent sums and their exponential bounds, then the spectral measure and compactification. Third establish convolution continuity in L^1 and the operator Taylor estimates; finite products should precede the infinite limit. Fourth formalize the two-point risk lower bound and the tensorization inequality. Finally add Dini's theorem, empirical moments, and the diagonal estimator for the compact-class criterion.

No declaration in the current repository is asserted to implement these new endpoints. The mathematical proof dependencies are explicit enough that one can formalize the layers independently rather than importing the conclusions as axioms.

15 Further research questions and topics

Research question 1 (Sharp minimax rates under spectral envelopes). Determine matching lower and upper rates for Gaussian-variance estimation under the two-sided envelopes (12.5) and (12.7). Are the logarithmic and stretched-logarithmic upper bounds proved here optimal, or does the integer-multiplicity constraint permit a substantially faster method?

The compactness criterion settles possibility, not speed. A lower bound must preserve the fixed envelope while producing moment or likelihood cancellation; the unrestricted equal-dust construction does not do this.

Research question 2 (Removing lower envelopes quantitatively). Obtain sharp rates assuming only $a_j \leq Cj^{-\beta}$ or $a_j \leq Cq^j$. Can positive polynomials adapted to a scale threshold replace the explicit root polynomial without incurring its coefficient growth?

Uniform consistency already follows from theorem 8.1. What remains is an effective quantitative modulus that does not presume a positive lower bound for every leading squared scale.

Research question 3 (Optimal recovery from finitely many exact cumulants). Given $v, p_2, \dots, p_m$, characterize the largest and smallest possible Gaussian variances among actual uniform-factor models. Determine when a Christoffel upper bound is attainable subject to the discrete condition $\rho(\{x\})/x \in \mathbb{N}$ for every positive atom.

Classical truncated moment feasibility alone drops this condition. This is a natural place where the special convolution model may improve on a generic positive-measure relaxation.

Research question 4 (A stable computational implementation). Design a numerically efficient estimator with certified error bounds from noisy cumulants. Compare coefficient balls, orthogonal-polynomial bases, positive-semidefinite moment projection, and interval arithmetic. A useful implementation should report a noise-dependent degree rather than simply invert a large Hankel matrix.

The exact formulas in this article establish a reference target; the finite rational-grid construction establishes existence, not practical scalability.

Research question 5 (Perturbations of geometric spectra). For $x_j = Cr^j(1 + \varepsilon_j)$ with controlled ε_j , determine whether the exact geometric Christoffel asymptotic persists with a stable leading constant. Identify the weakest summability or regular-variation assumptions yielding a relative-error theorem.

A first route is determinant perturbation, but its conditioning may be much worse than that of the variational problem. Direct comparison of positive spectral measures may be more robust.

Research question 6 (Higher-order dust expansions). Extend theorem 5.1 to arbitrary fixed order with constants independent of rank. Express the remainder in a small collection of power sums and derive uniform relative errors when several leading cumulants have been matched.

The double-telescoping proof suggests a route via higher operator differences. The issue is controlling the remainder without an exponentially large subset expansion. Existing geometric Edgeworth reports address a different scaling regime and should be compared carefully before any novelty claim.

Research question 7 (The unsmoothed statistical experiment). Which topology and local risk statements survive when the known smooth background is removed, particularly at the point mass and at finite uniform sums? Derive sharp distinctions between weak, total-variation, and Hellinger behavior at support boundaries.

The smoothing assumption is used explicitly in the TV-homeomorphism proof. It must not be silently discarded, even though some dust approximations have stronger convergence without smoothing.

Research question 8 (Unknown or partly known background). What aspects of Gaussian variance remain identifiable when B is known only through finitely many moments, or belongs to a specified compact family? Formulate sufficient restrictions that distinguish background Gaussian content from the latent Gaussian component.

The present subtraction of background cumulants is exact. With an arbitrary unknown background, convolution factorization creates genuine nonidentifiability, not merely statistical instability.

Research question 9 (Multivariate segment convolutions). Replace $a_j U_j$ by $u_j U_j$ with $u_j \in \mathbb{R}^p$, and Gaussian variance by a positive semidefinite covariance matrix. Characterize the corresponding matrix-valued dust, directional cumulant measures, and compact-class learnability conditions.

In one dimension the ordered nonnegative spectrum removes sign and permutation ambiguity. In several dimensions, opposite directions and angular moment identification introduce additional structure that must be resolved first.

Research question 10 (A proof-assistant endpoint with statistical semantics). Formalize theorem 8.1 together with theorem 6.1 for the actual probability-law model, not only for finite algebraic moment data. Can the spectral-measure representation reduce the amount of special-function analysis needed in that formalization?

The finite Schur-complement and determinant lemmas give a manageable initial milestone, while the compact statistical experiment provides a substantial longer-term target directly aligned with ProveIt’s verification goals.

16 Conclusion

The important distinction is not simply finite versus infinite sums. It is whether a class permits a fixed amount of variance to migrate into arbitrarily many unresolved factors. In the full class, that migration produces exact constant minimax risks at every finite sample size. On compact subclasses, uniformly vanishing energy tails are exactly the condition that restores uniform Gaussian-variance estimation.

The variance-weighted spectral measure makes this distinction transparent: the Gaussian component is its mass at zero. Positive polynomial filters and Christoffel minimization recover this atom from higher cumulants. The exact geometric formula supplies a particularly explicit link back to the Fabius–Rvachev setting, and bounded-coefficient regularization translates the algebra into honest noise and sampling guarantees. The remaining open work is chiefly quantitative: sharp envelope rates, efficient certified algorithms, and extensions beyond the known smooth background.

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